

Methods Section (Concise)

Neural Representation of Color in Color Vision Deficiency

Draft for PI Review

Jinil Kim

April 4, 2026

Note: This concise version (target: 4-5 pages main + 3 pages supplementary) moves detailed equations and procedures to Supplementary Methods. Main methods focus on essential information for understanding the study design and analysis approach.

1 Data collection

Participants

Twelve volunteers (IRB #2510/002-023) completed the study. Color vision was assessed using the Ishihara test Ishihara, 1917: 7 healthy controls (HC; 3 females, age 22.7 ± 2.5 years) and 2 with color vision deficiency (CVD; 1 deuteranomaly, 1 protanomaly, both male). Three participants were excluded for incomplete data. Given the small CVD sample, results are interpreted as individual case demonstrations using Crawford and Howell (1998) single-case statistics.

fMRI experiment

Participants viewed eight colors (0° – 315° in CIE $L^*a^*b^*$ space Commission Internationale de L'Eclairage, 1986) plus gray, presented in a rapid serial visual presentation (RSVP) detection task Brouwer and Heeger, 2009. Each 1.5s stimulus was followed by letter strings; participants detected consecutive 'K's. Stimulus order was optimized using Neurodesign Durnez et al., 2018. ISIs varied (3–6s). Each run (7 min) included 8 repetitions per color across 6 runs per session. CVD participants completed two sessions: original stimuli and filtered conditions.

MRI data were acquired on a Siemens 3T scanner (24 oblique slices perpendicular to calcarine sulcus Ryu and Lee, 2024; 1.5s TR, 30ms TE, $2 \times 2 \times 2$ mm voxels).

2 Preprocessing and response estimation

Data were converted to BIDS format Levitas et al., 2024 and coregistered using mutual information optimization Maes et al., 1997; Wells et al., 1996 via FreeSurfer. T1w-to-MNI normalization used FLIRT Jenkinson et al., 2002 and FNIRT Andersson et al., 2007 (2mm resolution).

Bilateral V1, V2, V3, and hV4 were defined using the Wang atlas Wang et al., 2015 (50% probability threshold). Voxel counts: V1 655 ± 214 , V2 451 ± 145 , V3 103 ± 29 , hV4 63 ± 22 (mean $\pm$ SD).

Neural responses were estimated via finite impulse response (FIR) modeling Brouwer and Heeger, 2009; Dale, 1999 to extract ROI-specific HRFs (8 TRs), followed by GLM fitting with

HRF and temporal derivative. Top 50% of voxels by variance explained were retained (V1 328 ± 107 , V2 226 ± 73 , V3 52 ± 15 , hV4 32 ± 11). Responses were aligned across runs using Procrustes transformation Gower, 1975 without scaling, improving cross-run correlation from $r = 0.54$ to $r = 0.71$.

3 Shared representational space

Cross-subject alignment used shared response modeling Chen et al., 2015, projecting individual voxel patterns into a low-dimensional common space $S \in \mathbb{R}^{k \times c}$ where k is the reduced dimension and $c = 8$ colors. For participant i with voxel responses X_i :

$$X_i = W_i S + E_i, \quad \text{s.t.} \quad W_i^T W_i = I_k \quad (1)$$

The common space was estimated from HC participants only; CVD data were projected into this HC-derived space. To quantify HC-CVD disparity, we used leave-one-out (LOO) references: for each fold, both the held-out HC and CVD participants were compared to the mean of remaining six HC participants, ensuring unbiased comparisons. Significance was assessed via Crawford and Howell (1998) modified t -tests. Representational dissimilarity matrices (RDMs) used correlation distance; color pair deviations were tested via 95% bootstrap CIs (1,000 iterations).

Dimension k was selected via LOSO cross-validation maximizing split-half RDM reliability Cunningham and Yu, 2014: V1 $k = 4$ (0.60), V2 $k = 4$ (0.57), V3 $k = 3$ (0.55), hV4 $k = 3$ (0.32).

4 Forward encoding model

We modeled hue-selective responses using six half-wave rectified squared sinusoidal channels Brouwer and Heeger, 2009, 2013; Freeman and Adelson, 1991: $r_k(\theta) = \max(0, \cos(\theta - \theta_k))^2$. Voxel responses B were modeled as $B = WC$, where C are channel outputs and W is the weight matrix estimated via ridge regression with regularization parameter selected by generalized cross-validation Golub et al., 1979.

Three cross-validation schemes tested model generalization:

LORO (discrimination): Leave-one-run-out tested color discriminability. Weight matrix W was trained on 5 runs, tested on the held-out run.

LOCO (interpolation): Leave-one-color-out tested whether the model could interpolate to novel colors. W was trained on 7 colors, tested on the held-out color—assessing preservation of circular hue geometry.

LOSO (zero-shot): Leave-one-subject-out tested cross-subject generalization using SRM-aligned data.

Evaluation metrics: (1) mean absolute error (MAE; chance = 90°) and categorization accuracy (chance = 12.5%) for decoding; (2) Pearson correlation for voxel prediction. Significance via permutation tests (10,000 iterations; Nichols and Holmes, 2002).

5 Behavioral tasks

Participants completed same-different discrimination to estimate JND thresholds Levitt, 1971 using adaptive staircases (two per color pair, eight reversals). Eight color pairs were selected from RDM analysis: three differing in both CVD types, two per CVD type only, and one control. An 8-AFC color identification task assessed categorical perception.

6 Reproducibility

Analyses used Python 3.10 (numpy 1.24.3, scipy 1.11.3, scikit-learn 1.3.0, BrainIAK 0.11).
Code: https://github.com/haba6030/colorBlind_analysis. Data available upon request.

Supplementary Methods

S1. Preprocessing details

Confound regression: Head motion was estimated via rigid-body realignment. No temporal filtering or confound regression was applied; slow drift was modeled via linear per-run regressors in GLM.

Registration QC: ROI coverage averaged 84.3% (SD = 21.7%); GLM valid ratio 99.6%. Sub-07 showed 30.8% coverage due to anatomical variability but was retained.

Voxel counts: Detailed per-subject voxel counts in Table S1. Variability reflects individual anatomy and registration quality.

S2. Procrustes alignment details

Runs 2–6 were aligned to run 1 via orthogonal transformation $X_{\text{aligned}} = XQ$ where Q minimizes $\|X_{\text{ref}} - XQ\|_F$. Scaling was prohibited to preserve response magnitudes. Cross-run correlation (15 pairs) improved from 0.54 to 0.71.

S3. SRM mathematical details

SRM minimizes reconstruction error across participants:

$$\min_{W_i, S} \sum_{i=1}^N \|X_i - W_i S\|_F^2 \quad \text{s.t.} \quad W_i^T W_i = I_k \quad (2)$$

CVD projection: $W_{\text{CVD}} = UV^T$ where $USV^T = \text{SVD}(X_{\text{CVD}} \cdot S^\dagger)$. LOO procedure: for fold i , reference R_{-i} is mean of remaining six HC. Disparity: $d(\text{participant}, R_{-i})$ using correlation distance.

S4. Forward model equations

Channel outputs for hue θ : $C_k(\theta) = \max(0, \cos(\theta - \theta_k))^2$ for $k = 1, \dots, 6$.

Ridge regression: $\hat{W} = B_1 C_1^T (C_1 C_1^T + \alpha I)^{-1}$

GCV criterion: $\text{GCV}(\alpha) = \frac{1}{n} \|y - \hat{y}\|_2^2 / (1 - \text{tr}(H)/n)^2$ where $H = X(X^T X + \alpha I)^{-1} X^T$

Color decoding: $\hat{C} = (\hat{W}^T \hat{W})^{-1} \hat{W}^T B_2$; hue estimate via $\arg \min_{\theta} \|\hat{C} - C(\theta)\|_2$

Voxel prediction: $\hat{B} = \hat{W} C_2$

S5. SRM consistency validation

SRM was compared to PCA and standard Procrustes. SRM yielded higher cross-subject RDM correlations (mean = 0.51) vs. PCA (0.38) and Procrustes (0.42), confirming its suitability.

S6. Mean activation analysis

Mean response amplitudes across colors did not differ between HC and CVD (all $p > .10$), confirming that representational differences were independent of overall activation magnitude.

S7. Filter design

CVD distortion was modeled as cone-specific spectral shift ($\Delta\lambda$ in nm) applied to stimulus hue angles Brettel et al., 1997. Details pending Phase 2 completion.

S8. Statistical analysis

Significance threshold $\alpha = 0.05$ with FDR correction Benjamini and Hochberg, 1995 ($q = 0.05$) for RDM comparisons (28 pairs). Crawford-Howell modified t -test: $t^* = (x_{\text{CVD}} - \bar{x}_{\text{HC}}) / (\text{SD}_{\text{HC}} \times \sqrt{(n+1)/n})$, $df = 6$. Effect sizes via Hedges' g . Permutation tests: 10,000 iterations (group), 1,000 (color-level).

S9. Behavioral-neural concordance

Full concordance analysis pending behavioral data completion (current $n = 5$ HC, 2 CVD). Preliminary data suggest correspondence between LOCO performance and elevated JND thresholds.

References

- Andersson, J. L., Jenkinson, M., & Smith, S. (2007). *Non-linear registration aka spatial normalisation* (Technical Report No. TR07JA2). FMRIB.
- Benjamini, Y., & Hochberg, Y. (1995). Controlling the false discovery rate: A practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society: Series B (Methodological)*, 57(1), 289–300. <https://doi.org/10.1111/j.2517-6161.1995.tb02031.x>
- Brettel, H., Viénot, F., & Mollon, J. D. (1997). Computerized simulation of color appearance for dichromats. *Journal of the Optical Society of America A*, 14(10), 2647–2655. <https://doi.org/10.1364/JOSAA.14.002647>
- Brouwer, G. J., & Heeger, D. J. (2009). Decoding and reconstructing color from responses in human visual cortex. *Journal of Neuroscience*, 29(44), 13992–14003. <https://doi.org/10.1523/JNEUROSCI.3577-09.2009>
- Brouwer, G. J., & Heeger, D. J. (2013). Categorical clustering of the neural representation of color. *Journal of Neuroscience*, 33(39), 15454–15465. <https://doi.org/10.1523/JNEUROSCI.2472-13.2013>
- Chen, P. H., Chen, J., Yeshurun, Y., Hasson, U., Haxby, J., & Ramadge, P. J. (2015). A reduced-dimension fmri shared response model. *Advances in Neural Information Processing Systems*, 28, 460–468.
- Commission Internationale de L’Eclairage. (1986). *Colorimetry* (2nd) [Publication No. 15.2]. CIE.
- Crawford, J. R., & Howell, D. C. (1998). Comparing an individual’s test score against norms derived from small samples. *The Clinical Neuropsychologist*, 12(4), 482–486. <https://doi.org/10.1076/clin.12.4.482.7241>
- Cunningham, J. P., & Yu, B. M. (2014). Dimensionality reduction for large-scale neural recordings. *Nature Neuroscience*, 17(11), 1500–1509. <https://doi.org/10.1038/nn.3776>
- Dale, A. M. (1999). Optimal experimental design for event-related fmri. *Human Brain Mapping*, 8(2-3), 109–114. [https://doi.org/10.1002/\(SICI\)1097-0193\(1999\)8:2<109::AID-HBM7>3.0.CO;2-W](https://doi.org/10.1002/(SICI)1097-0193(1999)8:2<109::AID-HBM7>3.0.CO;2-W)
- Durnez, J., Moerkerke, B., & Nichols, T. E. (2018). Neurodesign: Optimal experimental designs for task fmri. *Journal of Open Source Software*, 3(32), 564. <https://doi.org/10.21105/joss.00564>
- Freeman, W. T., & Adelson, E. H. (1991). The design and use of steerable filters. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 13(9), 891–906. <https://doi.org/10.1109/34.93808>
- Golub, G. H., Heath, M., & Wahba, G. (1979). Generalized cross-validation as a method for choosing a good ridge parameter. *Technometrics*, 21(2), 215–223. <https://doi.org/10.1080/00401706.1979.10489751>
- Gower, J. C. (1975). Generalized procrustes analysis. *Psychometrika*, 40(1), 33–51. <https://doi.org/10.1007/BF02291478>
- Ishihara, S. (1917). *Tests for colour-blindness*. Kanehara Shuppan.
- Jenkinson, M., Bannister, P., Brady, M., & Smith, S. (2002). Improved optimization for the robust and accurate linear registration and motion correction of brain images. *NeuroImage*, 17(2), 825–841. [https://doi.org/10.1016/s1053-8119\(02\)91132-8](https://doi.org/10.1016/s1053-8119(02)91132-8)
- Levitas, D., Hayashi, S., Vinci-Booher, S., Heinsfeld, A., Bhatia, D., Lee, N., Galassi, A., Niso, G., & Pestilli, F. (2024). Ez bids: Guided standardization of neuroimaging data interoperable with major data archives and platforms. *Scientific Data*, 11, 179. <https://doi.org/10.1038/s41597-024-02959-0>
- Levitt, H. (1971). Transformed up-down methods in psychoacoustics. *The Journal of the Acoustical Society of America*, 49(2), 467–477. <https://doi.org/10.1121/1.1912375>

- Maes, F., Collignon, A., Vandermeulen, D., Marchal, G., & Suetens, P. (1997). Multimodal-ity image registration by maximization of mutual information. *IEEE Transactions on Medical Imaging*, 16(2), 187–198. <https://doi.org/10.1109/42.563664>
- Nichols, T. E., & Holmes, A. P. (2002). Nonparametric permutation tests for functional neuroimaging: A primer with examples. *Human Brain Mapping*, 15(1), 1–25. <https://doi.org/10.1002/hbm.1058>
- Ryu, J., & Lee, S.-H. (2024). Slice planning perpendicular to calcarine sulcus for retinotopic mapping [In preparation]. *Journal of Neuroscience Methods*.
- Wang, L., Mruczek, R. E., Arcaro, M. J., & Kastner, S. (2015). Probabilistic maps of visual topography in human cortex. *Cerebral Cortex*, 25(10), 3911–3931. <https://doi.org/10.1093/cercor/bhu277>
- Wells, W. M., Viola, P. A., Atsumi, H., Nakajima, S., & Kikinis, R. (1996). Multi-modal volume registration by maximization of mutual information. *Medical Image Analysis*, 1(1), 35–51. [https://doi.org/10.1016/S1361-8415\(01\)80004-9](https://doi.org/10.1016/S1361-8415(01)80004-9)